STAT 709 Mathematical Statistics Lecture 14 - Fall 2026

Lecture 14: Concentration inequalities

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Priority guide

High	Core concepts to master for this lecture; will be directly examined.
Mid	Supporting concepts useful in other lectures; in scope of examinations.
Low	Enrichment supporting understanding; will not be examined.
Further reading	Optional self-study, references, extensions, and historical notes.

High priority *Overview*

A limit theorem describes a distribution as the sample size grows. A concentration inequality instead controls a probability at the sample size in hand. The central tools are the exponential Markov bound, the **subgaussian** moment-generating-function bound, and Bernstein's two deviation scales. We track which assumptions justify each bound and how to turn a tail probability into a confidence radius.

High priority *Introduction*

In this lecture, we will introduce some basics for understanding high-dimensional phenomena.

What are concentration inequalities? From the LLN, we know that an empirical mean $\widehat{\mu}$ "concentrates" around the population mean μ. This is characterized by the asymptotic notion of convergence in probability. When we zoom in on the difference $\widehat{\mu} - \mu$ by multiplying it with a factor of $\sqrt{n}$, then, under the i.i.d. finite-variance CLT assumptions, $\sqrt{n}(\widehat{\mu} - \mu)$ has a normal distribution asymptotically.

Concentration inequalities provide non-asymptotic bounds on the difference as an alternative way of characterizing the concentration. Usually we will see statements like "With very high probability, a certain event (e.g. inequality) happens." Often, the exception probability decays exponentially in the number of variables. Compared with classical asymptotic theory, concentration inequalities often give us refined quantitative control of extreme/exception probabilities.

In this lecture, we use C and c for positive absolute constants, which may change from line to line. An unsubscripted vector norm $\|a\|$ means $\|a\|_2$.

1 Motivation for concentration inequalities

Mid priority *Motivation for concentration inequalities*

Recall the central limit theorem: if $X_1, X_2, \dots X_n \overset{\text{i.i.d.}}{\sim} X$ satisfy $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$, then

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \xrightarrow{d} \mathcal{N}(0, 1).$$

Recall the Gaussian tail probability bounds from Lecture 1.

Proposition 1 (Gaussian tail probability)**.** *Let $G \sim \mathcal{N}(0,1)$, then for any $t > 0$,*

$$\left(\frac{1}{t} - \frac{1}{t^3}\right)\frac{1}{\sqrt{2\pi}}e^{-t^2/2} \le \mathbb{P}(G \ge t) \le \frac{1}{t}\frac{1}{\sqrt{2\pi}}e^{-t^2/2}.$$

Let Φ and ϕ be the standard Gaussian CDF and density, respectively. From the above proposition, if $t \to \infty$, then

$$1 - \Phi(t) = (1 + o(1))\frac{\phi(t)}{t}.$$

If we pretend that the following approximation is valid,

$$\mathbb{P}\big(\frac{1}{n}\sum_{i=1}^{n} X_i > t\big) = \mathbb{P}\big(\frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i > \sqrt{n}\,t\big) \approx \mathbb{P}(G > \sqrt{n}t),$$

then the tail probability of empirical mean of n i.i.d. random variables is $O\left(e^{-nt^2/2}\right)$. The CLT alone does not justify this approximation: its convergence at fixed thresholds gives no relative-error control at the moving threshold $\sqrt{n}\,t$. Exponentially decaying bounds do hold under stronger tail assumptions, which we develop below.

Proposition 2 (Berry–Esseen inequality)**.** *Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. random variables with zero mean, variance one and finite absolute third moment. Then, for all $t \in \mathbb{R}$,*

$$\left|\mathbb{P}\left(\frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i \ge t\right) - \mathbb{P}(G \ge t)\right| \le \frac{C\mathbb{E}|X|^3}{\sqrt{n}},$$

where C is a positive constant.

The error term is $O\left(\frac{1}{\sqrt{n}}\right)$, much larger than $O\left(e^{-nt^2/2}\right)$ for a fixed $t > 0$ as $n \to \infty$. Moreover, the error term cannot be made smaller in general due to the next simple example.

Thus Berry–Esseen gives a valid additive bound for $\mathbb{P}(n^{-1}\sum_i X_i \ge t)$, but its $n^{-1/2}$ error can be much larger than the exponentially small probability we want to bound.

Example 1. *Let $X_1, X_2, \ldots X_n \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(\frac{1}{2})$, where n is an even integer, then*

$$\mathbb{P}\left(\sum_{i=1}^{n} X_i = \frac{n}{2}\right) = 2^{-n}\binom{n}{n/2} = 2^{-n}\frac{n!}{[(n/2)!]^2}.$$

Applying Stirling's approximation:

$$n! = (1 + o(1))\sqrt{2\pi n}\left(\frac{n}{e}\right)^n,$$

we have

$$\mathbb{P}\left(\sum_{i=1}^{n} X_i = \frac{n}{2}\right) = (1 + o(1))\sqrt{\frac{2}{\pi}}\frac{1}{\sqrt{n}}$$

such that

$$\mathbb{P}\left(\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(X_i - \mathbb{E}X_i) = 0\right) = (1 + o(1))\sqrt{\frac{2}{\pi}}\frac{1}{\sqrt{n}}.$$

Note that $\mathbb{P}(G = 0) = 0$. In this example, the jump has order $n^{-1/2}$, so any continuous CDF differs from this discrete CDF on at least one side of the jump by half that size. After standardizing the Bernoulli variance, Berry–Esseen gives a matching upper order.

Mid priority *Practice: A continuous CDF cannot match a jump*

Exercise prompt. If a CDF F has a jump of size q at x_0 and G is continuous there, prove $\sup_x |F(x) - G(x)| \geq q/2$.

See Exercise 1 in Section 7 for the full statement and complete solution.

2 Subgaussian variables and Hoeffding's Inequality

High priority *Subgaussian variables and Hoeffding's Inequality*

To achieve our goal of bounding tail probabilities, we will need to build the machinery from scratch. First, we introduce an important definition.

Theorem 1 (Subgaussian random variables)**.** *Let X be a random variable with $\mathbb{E}X = 0$. Consider three conditions.*

(i) $\mathbb{P}(|X| \geq t) \leq 2e^{-t^2/K_1^2},\ \forall\, t \geq 0.$

(ii) $\sup_{p\geq 1} \frac{1}{\sqrt{p}}(\mathbb{E}|X|^p)^{1/p} \leq K_2.$

(iii) $\mathbb{E}[e^{\lambda X}] \leq e^{K_3^2\lambda^2},\ \forall\, \lambda \in \mathbb{R}.$[1]

The three conditions are equivalent up to absolute constants: if one holds with constant K_j, each other condition holds with a suitable $K_i \leq CK_j$. We say X is a subgaussian random variable if any of the three conditions hold.

Proof. (i)⇒(ii): Without loss of generality, assume $K_1 = 1$. Otherwise set $X := X/K_1$. Observe

$$\begin{aligned}
\mathbb{E}|X|^p &= \int_0^\infty \mathbb{P}(|X| \geq t) p t^{p-1} dt \qquad \text{(see Lecture 4, or use Fubini's theorem)} \\
&\leq \int_0^\infty 2pt^{p-1}e^{-t^2}\,dt \\
&= p\int_0^\infty u^{(p-2)/2}e^{-u}\,du \qquad \text{(change of variables } u = t^2\text{)} \\
&= p\,\Gamma(p/2) \qquad \text{(Gamma function)} \\
&\leq 3p(\frac{p}{2})^{p/2} \qquad \text{(by property of Gamma functions).}
\end{aligned}$$

(ii)⇒(iii): Let $\widetilde{X}$ be an independent copy, and $Z = X - \widetilde{X}$. Exchanging the copies shows that Z is symmetric. Jensen and independence give $\mathbb{E}\,e^{\lambda X} \leq \mathbb{E}\,e^{\lambda Z}$. From (ii) and Minkowski, $(\mathbb{E}\,|Z|^{2m})^{1/(2m)} \leq 2K_2\sqrt{2m}$. Symmetry, the nonnegative series for cosh, and Tonelli give

$$\mathbb{E}\,e^{\lambda Z} = \mathbb{E}\cosh(\lambda Z) = 1 + \sum_{m\geq 1} \frac{\lambda^{2m}\,\mathbb{E}\,Z^{2m}}{(2m)!}.$$

Using $(2m)! \geq (2m/e)^{2m}$ and $m! \leq m^m$, the m-th coefficient is at most $(2e^2K_2^2\lambda^2)^m/m!$. Therefore $\mathbb{E}\,e^{\lambda X} \leq \exp(2e^2K_2^2\lambda^2)$. The finite bound also justifies exponential integrability; Tonelli applies to nonnegative terms.

(iii)⇒(i): Without loss of generality, assume $K_3 = 1$. For any $\lambda > 0$, $t > 0$,

$$\begin{aligned}
\mathbb{P}(X \geq t) &\leq \mathbb{P}(e^{\lambda X} \geq e^{\lambda t}) \\
&\leq e^{-\lambda t}\mathbb{E}e^{\lambda X} \qquad \text{(by Markov's inequality)} \\
&\leq e^{-\lambda t + \lambda^2} \\
&\leq e^{-t^2/4} \qquad \text{(choose } \lambda = t/2 \text{ to minimize } -\lambda t + \lambda^2\text{)}.
\end{aligned}$$

[1] Note: $\mathbb{E}[e^{\lambda X}]$ is the moment-generating function (MGF) of X.

Repeating the same for $-X$, we have

$$\mathbb{P}(-X \geq t) \leq e^{-t^2/4}.$$

Therefore,

$$\mathbb{P}(|X| \geq t) \leq 2e^{-t^2/4}, \qquad \forall\, t > 0.$$

□

Taking pth roots in (i)⇒(ii) gives $\|X\|_{L^p} \leq C\sqrt{p}$, since $(3p)^{1/p}$ is uniformly bounded for $p \geq 1$.

The smallest K_2 in condition (ii) is called the **subgaussian norm** of X, denoted by $\|X\|_{\psi_2}$. By Minkowski inequality, it is easy to check that it satisfies the requirements of a norm (see Exercise 5). Like standard deviation, it measures the spread of a centered variable, but finite subgaussian norm additionally controls the tails. We use the moment definition for variables with any mean; only the uncentered MGF bound (iii) requires mean zero. Some textbooks use an exponential-moment definition of $\|X\|_{\psi_2}$; these conventions agree up to absolute constants.

Example 2. *If $X \sim \mathcal{N}(0, \sigma^2)$, then $\|X\|_{\psi_2} \leq C\sigma$.*

Example 3. *If $|X| \leq M$ a.s., then $\|X\|_{\psi_2} \leq CM$.*

For a sum of independent subgaussian random variables, we have the desired Gaussian tail bound.

Theorem 2 (Hoeffding's Inequality)**.** *Let $X_1, X_2, \ldots, X_n$ be independent random variables, such that $\mathbb{E}X_i = 0$ and each X_i is subgaussian. Let $a = (a_1, \ldots, a_n)$ be a nonrandom vector. Denote $K := \max_{1 \leq i \leq n} \|X_i\|_{\psi_2}$. Then*

$$\mathbb{P}\left(\left|\sum_{i=1}^n a_i X_i\right| \geq t\right) \leq 2\exp\left(-\frac{ct^2}{K^2\|a\|^2}\right) \tag{1}$$

holds for $t > 0$ and a positive absolute constant c. If $K\|a\|_2 = 0$, the sum is zero almost surely, so this degenerate case is handled directly.

In particular, if we choose $a_i = 1/n$ for all i then Hoeffding's inequality says that we have the Gaussian tail bound.

Apart from the constant, this bound cannot be improved. In the case $X_i \overset{\text{i.i.d.}}{\sim} N(0,1)$, we have $\sum_{i=1}^n a_i X_i \sim N(0, \|a\|^2)$, so by Proposition 1 its tail has exponent $-t^2/(2\|a\|^2)$ up to the stated polynomial factor.

Proof of Theorem 2.

$$\begin{aligned}
\mathbb{E}\exp\left(\lambda \sum_{i=1}^n a_i X_i\right) &= \prod_{i=1}^n \mathbb{E}\exp(\lambda a_i X_i) \qquad \text{(by independence)} \\
&\leq \prod_{i=1}^n \exp(C\lambda^2 a_i^2 \|X_i\|_{\psi_2}^2) \qquad \text{(by Theorem 1 (iii))} \\
&\leq \exp(C\lambda^2 \sum_{i=1}^n a_i^2 K^2) \\
&= \exp(C\lambda^2 \|a\|^2 K^2).
\end{aligned}$$

Then use Theorem 1 (i). □

Subgaussian scaling. A centered subgaussian variable has a Gaussian-type MGF bound. For independent weighted sums, the squared coefficient scales add; a Chernoff argument converts that sum into a tail bound.

High priority *Practice: Unequal weights and concentration*

Exercise prompt. For independent mean-zero variables with subgaussian norm at most K, compare equal weights $a_i = 1/n$ with weights $a_1 = 1$, $a_i = 0$ for $i > 1$ in Hoeffding's inequality. See Exercise 2 in Section 7 for the full statement and complete solution.

3 Subexponential variables and Bernstein's Inequality

High priority *Subexponential variables and Bernstein's Inequality*

We introduce another notion that characterizes the fast decay of the tail probability, though not as fast as subgaussian tails.

Theorem 3 (Subexponential random variables)**.** *Let X be a random variable with $\mathbb{E}X = 0$.*

(i) $\mathbb{P}(|X| \geq t) \leq 2e^{-t/K_1}$, $\forall\, t \geq 0$.

(ii) $\sup_{p\geq 1} \frac{1}{p}(\mathbb{E}|X|^p)^{1/p} \leq K_2$.

(iii) $\mathbb{E}[e^{\lambda X}] \leq e^{K_3^2\lambda^2}$, $\forall\, |\lambda| \leq 1/K_3$.

The three conditions are equivalent up to absolute constants: each condition with K_j implies the others with suitable $K_i \leq CK_j$. See [1, Proposition 2.7.1] for the parallel proof.

The smallest K_2 in condition (ii) is called the **subexponential norm** of X, denoted by $\|X\|_{\psi_1}$. By definition, subgaussian random variables are necessarily subexponential random variables with $\|X\|_{\psi_1} \leq \|X\|_{\psi_2}$; this is because

$$\frac{1}{p}(\mathbb{E}|X|^p)^{1/p} \leq \frac{1}{\sqrt{p}}(\mathbb{E}|X|^p)^{1/p}, \qquad \text{for any } p \geq 1$$

so taking $\sup_{p\geq 1}$ yields $\|X\|_{\psi_1} \leq \|X\|_{\psi_2}$. In fact, intuitively, a subexponential random variable can be viewed as the square of a subgaussian random variable. This intuition is captured by the following lemma.

Centering. For random variables with nonzero mean, conditions *(i)* and *(ii)* in Theorem 3 (and similarly in Theorem 1) are still equivalent. Moreover, we have bound

$$\|X - \mathbb{E}\,X\|_{\psi_2} \leq C\|X\|_{\psi_2}.$$

See Lemma 2.6.8 of the book [1].

Lemma 1. *Suppose that X, Y are subgaussian (not necessarily independent). Then XY is subexponential with*

$$\|XY\|_{\psi_1} \leq 2\|X\|_{\psi_2}\|Y\|_{\psi_2}.$$

Proof. We use the second condition of subgaussian and subexponential random variables.

$$\begin{aligned}[\mathbb{E}(|X|^p|Y|^p)]^{1/p} &\leq \left[\mathbb{E}|X|^{2p}\mathbb{E}|Y|^{2p}\right]^{1/(2p)} \qquad \text{(Cauchy-Schwarz Inequality)}\\ &\leq \|X\|_{\psi_2}\sqrt{2p}\,\|Y\|_{\psi_2}\sqrt{2p} = 2p\|X\|_{\psi_2}\|Y\|_{\psi_2}.\end{aligned}$$

□

Example 4. *Let $G \sim \mathcal{N}(0,1)$, then G^2 (namely chi-square χ_1^2) is subexponential, $\|G^2\|_{\psi_1} \leq C$.*

Example 5. *Let $X \sim Exp(1)$ be an exponential random variable, then $\|X\|_{\psi_1} \leq C$.*

Example 6. *Let X be a random variable such that $|X| \leq M$ is bounded for $M \geq 0$. Then, $\|X\|_{\psi_1} \leq CM$.*

Now we are in the position to state the concentration inequality for a sum of subexponential random variables.

Theorem 4 (Bernstein's Inequality)**.** *Let $X_1, \ldots, X_n$ be independent random variables, such that $\mathbb{E}X_i = 0$ and each X_i is subexponential. Let $a = (a_1, \ldots, a_n)$ be a nonrandom vector. Denote $K := \max_{1\le i\le n} \|X_i\|_{\psi_1}$. Then*

$$\mathbb{P}\left(\left|\sum_{i=1}^n a_i X_i\right| \ge t\right) \le 2\exp\left(-c\min\left(\frac{t^2}{K^2\|a\|^2}, \frac{t}{K\|a\|_\infty}\right)\right), \tag{2}$$

for $t > 0$, with a positive absolute constant c; a zero scale again means the sum vanishes almost surely.

Subexponential random variables have "heavier" tails compared with subgaussian random variables. Expectedly, the upper bound in Eqn. 2 is worse than the bound in Hoeffding's theorem, since it involves an additional term $t/(K\|a\|_\infty)$ in the exponent.

Proof. We apply a similar strategy as in the proof for Hoeffding's Inequality.

$$\mathbb{P}\left(\sum_{i=1}^n a_i X_i \ge t\right) \le e^{-\lambda t}\prod_{i=1}^n \mathbb{E}e^{a_i\lambda X_i}.$$

If $|\lambda| \le \frac{C}{\|a\|_\infty K}$, then by (iii) and the same argument for the negative sum,

$$\mathbb{P}\left(\left|\sum_{i=1}^n a_i X_i\right| \ge t\right) \le 2\exp\left(-\lambda t + C\lambda^2 K^2\|a\|^2\right).$$

Write the exponent as $-\lambda t + A\lambda^2$, where $A = CK^2\|a\|_2^2$, and let $L = c_0/(K\|a\|_\infty)$ be the permitted MGF range. For a nonzero scale, choose

$$\lambda = \min\{t/(2A), L/2\}.$$

If $t \le AL$, the exponent is $-t^2/(4A)$. If $t > AL$, it is at most $-Lt/2 + AL^2/4 \le -Lt/4$. These are the quadratic and linear terms in the claimed bound. If the sum is identically zero, handle it directly. □

Theorem 5 (Bernstein's Inequality for bounded random variables)**.** *Let $X_1, \ldots, X_n$ be independent random variables, such that $\mathbb{E}X_i = 0$ and $\max_{1\le i\le n}|X_i| \le M$ a.s.. Then*

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \ge t\right) \le 2\exp\left(-\frac{t^2/2}{\sigma^2 + Mt/3}\right),$$

where $\sigma^2 = \sum_{i=1}^n \mathbb{E}X_i^2$ is total variance.

Note that

$$\max\{\sigma^2, Mt/3\} \le \sigma^2 + Mt/3 \le 2\max\{\sigma^2, Mt/3\},$$

so the exponent in the upper bound of Theorem 5 is roughly (up to a constant[2])

$$-\frac{t^2}{\max\{\sigma^2, Mt\}} = -\min\left\{\frac{t^2}{\sigma^2}, \frac{t}{M}\right\}$$

This is similar to the exponent in the bound in (2)

[2]When we say two quantities s and t are up to a constant, we mean $ct \le s \le Ct$ for absolute constants $c, C > 0$.

Remark 1. *There is a popular variant of Bernstein's inequality.*

$$\mathbb{P}\left(|\sum_{i=1}^{n} X_i| \geq \frac{C(\sigma+M)}{\delta}\right) \leq 2\exp\left(-\min\left(\frac{1}{\delta}, \frac{1}{\delta^2}\right)\right), \qquad \text{for any } \delta > 0, \tag{3}$$

where C is a certain constant. This inequality is a consequence of Theorem 5. In fact, note that

$$\frac{-t^2/2}{\sigma^2 + Mt/3} \leq \frac{-t^2}{4\max\{\sigma^2, Mt/3\}} = -\min\left\{\frac{t^2}{4\sigma^2}, \frac{3t}{4M}\right\}.$$

So choosing $t = \frac{C(\sigma+M)}{\delta}$ with a sufficiently large C leads to

$$\frac{t^2}{4\sigma^2} \geq \frac{1}{\delta^2}, \qquad \frac{3t}{4M} \geq \frac{1}{\delta}$$

and therefore leads to the upper bound in Eqn. 3.

Instructor's Note *Two costs of a large deviation*

A moderate fluctuation can be built from many small contributions, so a variance scale is natural. A very large fluctuation can be driven by a large individual contribution, so the largest weight enters too. Bernstein's minimum in the exponent keeps both possibilities in view.

Two Bernstein regimes. Subexponential sums have a quadratic exponent for moderate deviations and a linear exponent for large deviations. Products of subgaussians naturally have this weaker tail behavior.

High priority *Practice: Use a Bernstein radius*

Exercise prompt. Let independent centered Y_i satisfy $|Y_i| \leq 2$ and $\sum_i \mathrm{Var}(Y_i) = 9$. Use Bernstein to give a radius with two-sided failure probability at most $2e^{-4}$.

See Exercise 3 in Section 7 for the full statement and complete solution.

4 Understanding the two tail probabilities

Mid priority *Understanding the two tail probabilities*

Let $X_1, X_2, \ldots, X_n$ be independent random variables, and $a = (a_1, \ldots, a_n)$ a nonrandom vector. We are interested in the tail probability of the sum $S_n = \sum_{i=1}^n a_i X_i$. When each X_i has mean zero and variance one, $\mathrm{Var}(S_n) = \|a\|^2$; use this normalization in the following comparison, with uniformly bounded tail norms.

1. **Gaussian distribution**. In the canonical case $X_i \overset{d}{=} N(0,1)$, we have $S_n \overset{d}{=} N(0, \|a\|^2)$, so the distribution of S_n is completely determined by its variance. According to Proposition 1, the tail probability is mainly determined by $e^{-t^2/(2\|a\|^2)}$ (ignoring poly-factor).

2. **Asymptotic normal distribution**. Under certain conditions we expect the asymptotic normality $S_n/\|a\| \overset{d}{\to} N(0,1)$, so approximately $S_n \overset{d}{\approx} N(0, \|a\|^2)$. However, as we explained in Section 1, the CLT controls fixed standardized thresholds; it does not by itself give a relative approximation for increasingly rare tails.

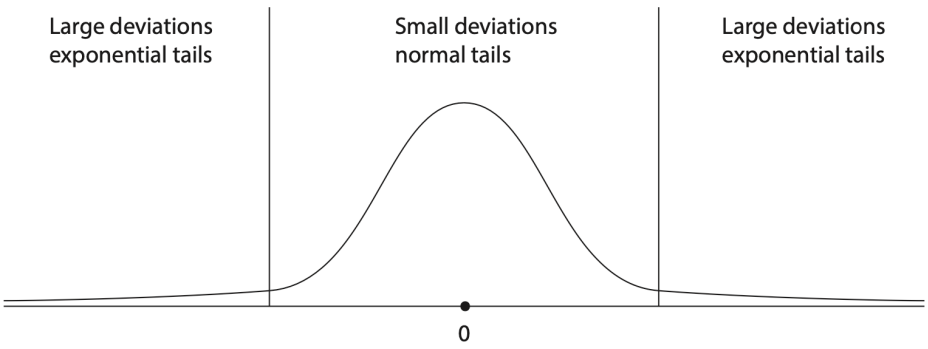


Figure 2.3 Bernstein's inequality for a sum of sub-exponential random variables gives a mixture of two tails: sub-gaussian for small deviations and sub-exponential for large deviations.

3. **Subgaussian distribution**. If the tail probability of each individual X_i is "lighter" (in the sense of Theorem 1(i)) than the standard Gaussian tail, then we still get a "light" tail controlled by $e^{-ct^2/(2\|a\|^2)}$ (up to constant c in the exponent).

4. **Subexponential distribution**. If the tail of X_i is slightly heavier than Gaussian, then S_n has an upper bound with two deviation regimes in Eqn. 2—one is Gaussian $e^{-ct^2/(2\|a\|^2)}$ and another is exponential $e^{-ct/\|a\|_\infty}$. The tail probability is controlled by the Gaussian tail when t is moderately large and by the exponential tail when t is very large. A Gaussian bound is unavailable in general: consider $a = (1, 0, \ldots, 0)$ and $X_1 \stackrel{d}{=} \text{Exp}(1) - 1$, where the tail probability is purely exponential.

Bernstein's bound has two scales: the accumulated tail scale $K\|a\|_2$ and the largest individual scale $K\|a\|_\infty$. The former gives a quadratic exponent for moderate deviations; the latter gives a linear exponent for large deviations. For independent unit-variance summands, $\|a\|_2$ is the actual standard deviation, but in general $K\|a\|_2$ is only a tail proxy. The accompanying figure illustrates the two regimes.

Theorem 5 controls the tail probability from a slightly different way: first using a variance quantity σ^2 similar to $\|a\|^2$ above, second using a crude bound M on X_i (trivially leads to subexponential norm M) to control the potentially "heavier" tail of X_i.

High-probability bound. Finally, to get a high-probability bound on S_n, we conclude that $|S_n|$ is bounded by the following quantity with probability at least $1 - \delta$.

1. Gaussian case: $C\sigma\sqrt{\log(2/\delta)}$, where $\sigma^2 = \text{Var}(S_n)$. For subgaussian summands, replace σ by the tail scale $K\|a\|_2$; this need not be comparable to the actual standard deviation.

2. Subexponential case: $CK[\|a\|_2\sqrt{\log(2/\delta)} + \|a\|_\infty \log(2/\delta)]$. For bounded summands, the variance-sensitive radius below is sharper than the coarser $C(\sigma + M)\log(2/\delta)$ form when $0 < \delta \le 1/2$.

A sharper explicit radius. For bounded centered summands, let $u = \log(2/\alpha)$ with $0 < \alpha < 1$. A useful Bernstein radius is

$$r_B(\alpha) = \sqrt{2\sigma^2 u} + \frac{2M}{3}u, \qquad \mathbb{P}(|S_n| \ge r_B(\alpha)) \le \alpha.$$

The square-root term represents accumulated variance; the linear term records the size of one bounded summand. For $\sigma^2 = 1$, $M = 1$, $u = 1$ and $u = 4$, the radii are $\sqrt{2} + 2/3$ and $\sqrt{8} + 8/3$.

Turn a tail bound into an error bar. For centered independent bounded summands, a sufficient Bernstein radius at failure probability α is $\sqrt{2\sigma^2 \log(2/\alpha)} + (2M/3)\log(2/\alpha)$, with total variance σ^2 and bound M.

Mid priority *Practice: Invert the bounded Bernstein bound*

Exercise prompt. Prove the explicit radius $r_B(\alpha)$ displayed above, assuming $\sigma^2, M > 0$.

See Exercise 4 in Section 7 for the full statement and complete solution.

5 Further reading

Further reading *Further reading*

The recent book [1] on high-dimensional probability provides an excellent introduction to this topic. The concentration material and the inherited deviation sketch are adapted from Chapter 2 of this book. The course uses its moment-norm convention, so numerical constants may differ from the book's exponential-moment norm. The author's first-edition text contains the cited numbering.

6 Key ideas to retain

High priority *Subgaussian scaling*

A centered subgaussian variable has a Gaussian-type MGF bound. For independent weighted sums, the squared coefficient scales add; a Chernoff argument converts that sum into a tail bound. See the source checkpoint.

High priority *Two Bernstein regimes*

Subexponential sums have a quadratic exponent for moderate deviations and a linear exponent for large deviations. Products of subgaussians naturally have this weaker tail behavior. See the source checkpoint.

Mid priority *Turn a tail bound into an error bar*

For centered independent bounded summands, a sufficient Bernstein radius at failure probability α is $\sqrt{2\sigma^2 \log(2/\alpha)} + (2M/3)\log(2/\alpha)$, with total variance σ^2 and bound M. See the source checkpoint.

7 Exercises and complete solutions

Mid priority *A continuous CDF cannot match a jump*

Exercise 1 (A continuous CDF cannot match a jump).
If a CDF F has a jump of size q at x_0 and G is continuous there, prove $\sup_x |F(x) - G(x)| \geq q/2$.

Solution.
Let $a = F(x_0-)$ and $b = F(x_0)$, so $b - a = q$. Continuity gives $G(x_0-) = G(x_0) = g$. Since $q \leq |b-g| + |g-a|$, one of the two errors is at least $q/2$. The left error is attained as a one-sided limit and hence is bounded by the supremum.

□

High priority *Unequal weights and concentration*

Exercise 2 (Unequal weights and concentration).
For independent mean-zero variables with subgaussian norm at most K, compare equal weights $a_i = 1/n$ with weights $a_1 = 1$, $a_i = 0$ for $i > 1$ in Hoeffding's inequality.

Solution.
Equal weights give $\|a\|^2 = 1/n$ and the bound $2e^{-cnt^2/K^2}$. A single nonzero weight gives $\|a\|^2 = 1$ and $2e^{-ct^2/K^2}$. Averaging independent terms produces the sample-size improvement; merely having n observed variables does not.

□

High priority *Use a Bernstein radius*

Exercise 3 (Use a Bernstein radius).
Let independent centered Y_i satisfy $|Y_i| \le 2$ and $\sum_i \operatorname{Var}(Y_i) = 9$. Use Bernstein to give a radius with two-sided failure probability at most $2e^{-4}$.

Solution.
Take $u = 4$, $\sigma^2 = 9$ and $M = 2$. The sufficient radius is $\sqrt{2(9)(4)} + 2(2)(4)/3 = 6\sqrt{2} + 16/3$. Indeed this radius satisfies $t^2 \ge 2u(\sigma^2 + Mt/3)$, so Bernstein gives the required bound. The exact positive quadratic root would be slightly smaller.

□

Mid priority *Invert the bounded Bernstein bound*

Exercise 4 (Invert the bounded Bernstein bound).
Prove the explicit radius $r_B(\alpha)$ displayed above, assuming $\sigma^2, M > 0$.

Solution.
Let $a = \sqrt{2\sigma^2 u}$ and $b = 2Mu/3$. For $t = a + b$, $t^2 - (2uM/3)t - 2u\sigma^2 = (a+b)^2 - b(a+b) - a^2 = ab \ge 0$. Therefore $t^2/[2(\sigma^2 + Mt/3)] \ge u$ and the tail bound is at most $2e^{-u} = \alpha$. If all summands vanish the assertion is immediate.

□

High priority *The moment subgaussian norm*

Exercise 5 (The moment subgaussian norm).
Verify homogeneity, definiteness and the triangle inequality for $\|X\|_{\psi_2} = \sup_{p \ge 1} p^{-1/2} \|X\|_{L^p}$.

Solution.
Homogeneity follows from that of each L^p norm. If the supremum is zero, its $p = 1$ term gives $\mathbb{E}\,|X| = 0$, so $X = 0$ a.s.; the converse is immediate. Minkowski gives $p^{-1/2}\|X + Y\|_{L^p} \le p^{-1/2}\|X\|_{L^p} + p^{-1/2}\|Y\|_{L^p}$, and taking suprema proves the triangle inequality on the space where the norm is finite.

□

High priority *Optimize a quadratic MGF bound*

Exercise 6 (Optimize a quadratic MGF bound).
If $\mathbb{E}\, e^{\lambda Z} \le e^{v\lambda^2/2}$ for every real λ and $v > 0$, prove $\mathbb{P}(|Z| \ge t) \le 2e^{-t^2/(2v)}$.

Solution.

For $\lambda > 0$, Markov gives $\mathbb{P}(Z \geq t) \leq e^{-\lambda t + v\lambda^2/2}$. The minimum occurs at $\lambda = t/v$, giving $e^{-t^2/(2v)}$. Repeat for $-Z$ and use the union bound. At $t = 0$, the bound is trivially valid.

□

High priority *Centering a tail norm*

Exercise 7 (Centering a tail norm).
For either moment norm ψ_1 or ψ_2, show $\|X - \mathbb{E}\,X\|_{\psi_j} \leq 2\|X\|_{\psi_j}$.

Solution.
The norm of the constant $\mathbb{E}\,X$ is $|\,\mathbb{E}\,X|$, because the supremum of $p^{-1/j}$ is 1. Also $|\,\mathbb{E}\,X| \leq \mathbb{E}\,|X| \leq \|X\|_{\psi_j}$. The triangle inequality gives the claim.

□

Mid priority *Locate the two-scale crossover*

Exercise 8 (Locate the two-scale crossover).
For a nonzero weight vector, find where $t^2/(K^2\|a\|_2^2) = t/(K\|a\|_\infty)$ for $t > 0$. Which exponent is smaller on either side?

Solution.
Cancel t to obtain $t_* = K\|a\|_2^2/\|a\|_\infty$. For $0 < t < t_*$ the quadratic term is smaller; for $t > t_*$ the linear term is smaller. Equal weights $1/n$ give $t_* = K$, while the exponent in the quadratic range grows proportionally to n.

□

References

[1] Roman Vershynin. *High-dimensional probability: An introduction with applications in data science*, volume 47. Cambridge university press, 2018.